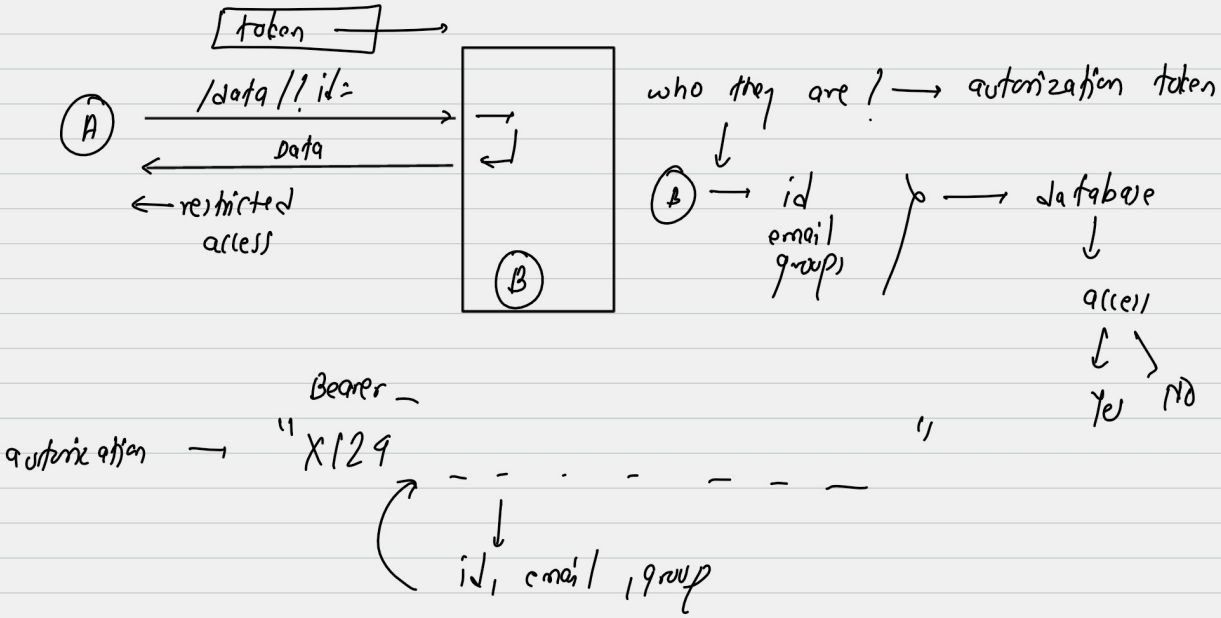
25-10-2026

Agenda: → Industry Use case

python → Exception-Handling, context-manager, File-operation

• ISO →





Mode	Meaning	pointer_start	wipe_file	usage	create_file ?
r	Read only	Beginning	NO	standard reading	NO
w	write only		Yes	fresh start, delete old data	Yes
a	append	End	NO	Add more content	Yes
r+	Read + write	Beginning	NO	modify existing content	NO
w+	write + read	Beginning	Yes	fresh start, delete old data	Yes
a+	Append + read	End	NO	Add more content + read ≠ modify	Yes

f. → reading → read() → full content
→ readline() → extract only one line at time
→ readlines() → [line-1, line-2, line-3]

→ writing → write()
→ writelines(line, line)

→ seek → shift the cursor or pointer
seek(0) → BOF

.tell()

ISO → International Organization for Standardization

Certification

ISO 9001
ISO 27000

Standard → how a company should build, manage & secure its software product.

→ software
→ food safety, Env mgement, health

ISO → 9001
→ ISO/IEC 27000 |
→ 20000-1
→ 42001

NOT: define coding rules

ISO - define the standard

- publishing requirement
- principles
- guidance for specific areas

client / regulator / company

Determines which standard or obligation apply to the product & project.

↓

Engineering & other teams

Translate applicable requirements into architecture, code, test, processes & records

↓

Evidence & audit

The company demonstrate that it follows the required practices and controls

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Standard	Main Question	Team to-do
ISO 9001 (Quality)	Can we deliver quality consistently?	Define requirement, review code, test changes, track defects & releases
ISO/IEC/IEEE 90003 (software quality guidance)	How can quality management practices apply to software?	Apply quality process to software development & maintaince.
ISO/IEC 27001 (Security)	How do we manage information security risk	control access protect secrecy & data manage vulnerabilities, document security risks.
ISO 26262 (Automotive functional safety)	could a software or system failure create a road vehicle safety hazard?	Analysis, trace safety requirement, verify software

Client ask for a ISO compliance?

Client
↓
AI-SAAS application

→ Client → our suppliers must meet our ISO/IEC 27001 security requirements.

→ clarify → exact scope
→ contract requirement
→ evidence expected

→ project manager + security / compliance lead

→ convert those req. into assigned task, policies, risk assessment, acceptance criteria.
→ identify the owner of each requirement.

→ Architect + tech lead

- Decide what the system needs:
  - authentication, role-base permission, encryption, secure architecture, audit log, so on

→ Developers
→ implement necessary code
→ infra changes
→ data-access in the api

→ QA / security team
→ Test the access restriction work
→ inspect vulnerabilities
→ verify behavior

→ project / compliance owner

→ Maintain the evidence: requirement records, code-review history, test reports, risk decision, approvals, release records.
: prepare for internal & external audit as applicable.